

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2026 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTEEC602 - ELECTRICAL MACHINE DESIGN

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 D.C .windings are
 - a. Some times 2-layer type
 - b. Never 2-layer type
 - c. Always 2-layer type
 - d. None of the above
- 2 Which of the following is an advantage of hydrogen cooling?
 - a. Increase in efficiency
 - b. Increase in ratings
 - c. Increase in life
 - d. All of the above
- 3 In a D.C. machine the current per brush arm should not be more than
 - a. 100 A
 - b. 200 A
 - c. 300 A
 - d. 400
- 4 The weight of copper of both armature and field windings in a DC machine generally decreases with:
 - a) Increase in speed
 - b) Increase in number of poles
 - c) Decrease in voltage
 - d) Decrease in armature current
- 5 To reduce Electrical energy bill, power factor should be kept
 - a. as less as possible
 - b. as high as possible
 - c. power factor does not affect energy bill
 - d. as close to unity as possible
- 6 Dummy coil should not be used in
 - a. Small machines
 - b. Large machines
 - c. Either (a) or (b)
 - d. None of the above
- 7 Machines cooled by direct cooling method may be called
 - a. Supercharged
 - b. Inner cooled
 - c. Conductor cooled
 - d. Any of the above
- 8 The stator of a D.C. machine comprises of
 - a. Main poles
 - b. Inter poles
 - c. Frame
 - d. All of the above
- 9 Which of the following materials is generally used in the design of electromagnetic machines?
 - a Paramagnetic materials
 - b Diamagnetic materials
 - c Ferromagnetic materials
 - d Core-magnetic materials

- 10 What kind of materials are used to design electrical heating devices?
 - a High Conductivity Materials b High Resistivity Materials
 - c High Weldability Materials d High Solderability Materials
- 11 What are the main dimensions of a rotating electric machine?
 - a Field diameter and Armature core length
 - b Armature diameter and Field core length
 - c Armature diameter and Armature core length
 - d Field diameter and Field core length
- 12 **Specific magnetic loading in electrical machine design is defined as:**
 - a) Total ampere conductors per meter
 - b) Flux per pole per unit armature area
 - c) Average flux density over the armature surface
 - d) Current density in the conductor

Q2. Solve the following.

- A) Explain introduction & advantages of CAD designing of Electrical machine. 6
- B) Write a short note on: i) Principle of Design ii) Ratings 6

Q3. Solve the following.

- A) A 250 V, 37 kw, d.c. shunt motor has to exert a maximum torque of 150 percent of 6 full load torque during the starting period. The resistance of armature circuit is 0.2 ohm and the full load efficiency is 84 percent. The number of studs is 8. Determine : i) the upper and lower limits of current during starting ii) the resistance of each section.
- B) Write a short note on: i) lifting magnets. ii) Chokes 6

Q4. Solve Any Two of the following.

- A) Derive the output equation of three phase induction motor. 6
- B) Find the main dimensions of a 15kw, 3 phase 400V, 50Hz, 2810 rpm squirrel cage 6 induction motor having an efficiency of 0.88 and a full load power factor of 0.9. Assume: Specific magnetic loading = 0.5 wb/m^2 . Specific electric loading = 25000 A/m. Take the rotor peripheral speed as approximately 20m/s at synchronous speed. <https://www.batuonline.com>
- C) Explain harmonic field effect on the performance of 3-phase induction motor 6

Q5. Solve Any Two of the following.

- A) Explain Heating and Cooling cycles 6
- B) Explain Temperature rise and heat dissipation 6
- C) Explain Study of different modes of heat generation 6

Q6. Solve Any Two of the following.

- A) Derive the output equation of single phase transformer 6
- B) Calculate the approximate overall dimensions for a 200 KVA, 6600/440 V, 50 Hz, 3 phase core type transformer. The following data may be assumed : emf per turn = 10 V, maximum flux density = 1.3 wb/m^2 , Current density = 2.5 A/mm^2 , window space factor = 0.3, overall height = overall width, stacking factor 0.9. Use a 3 stepped core. for a 3 stepped core : Width of largest stamping = $0.9 d$ and Net iron area = $0.6d^2$. Where d is the diameter of circumscribing circle.
- C) Explain in detail functions of breather and conservator. Also state their locations. 6